Presentation 4

Links to presentation(s) and code(s) on GitHub

Presentation:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/OutcomeTagging_Nov18_EDASimple2.pdf

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/VideoLink_Nov18_EDASimple2.txt

Code:

Data imported with these scripts:

CAR:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/CAR_data_import_R_CarolineCorr.R

Scripts that I worked on:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/OutcomeTagging_Nov17_CarolineCorr.R

What did you do?

I developed the method for tagging the outcomes through a summary table and case when logic and wrote the code to execute it in R. Hailee, Nikole, and myself then split up the ~90 last activities in the Excel sheet and wrote the case when statements to map all of their outcomes/sub-types.

How does it help the project?

This helped the project by giving us a clear, straightforward way to tag the outcomes with easy-to-understand case when logic. This approach allowed us to tag significantly more contact session IDs than our last attempt at outcome categorization because it focused on pulling the final value for different relevant fields (activity, ep, flow, etc) that the ID traversed to and summarizing the call journey into one row per contact session ID, making the aggregations very simple.

Issues faced (if any)

We didn't encounter any real issues while coding the tags, but we did have some questions on how to categorize a lot of positive outcomes that are related to Courtesy Callbacks and queues, for they end up in their own menu instead of relating back to one of the main 8 menus that we were asked to look at the outcomes for.

Attempts to resolve issues (if any)

Issues resolved (if any)

Next steps

We need to figure out a way to ensure that positive outcomes related to the main 8 menus are not lost through our current approach since they technically ended in a queue/courtesy callback, but that is not captured when only the last activity/flow/ep is pulled into the summary.

References (Mention if you built up on someone else's work)

Presentation 3

Links to presentation(s) and code(s) on GitHub

Presentation:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/CallMenuOutcomes_4Nov_EDATeam2.pdf

** Video was too large to push to GitHub – emailed to Krish instead and posted .txt with link

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/VideoLink_Nov5_EDASimple2.txt

Code:

Data imported with these scripts:

CAR:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/CAR_data_import_R_CarolineCorr.R

Scripts that I worked on:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/FamilyMenuOutcomes_Nov3_CarolineCorr.R

What did you do?

I wrote the code and created the table detailing the percentages for each outcome type (positive, negative, neutral) for the Family Menu. To do this, I collected the contact session IDs that traversed to this menu and ended in a relevant menu (that is, if a call went to the family menu but ended up in a completely separate menu before the call was terminated, it would be discarded). I then tagged outcome types by considering termination reason and less intuitive clues such as the last menu traversed to and whether or not LegalAid services it as well as if an agent was assigned to the call before terminating.

How does it help the project?

This helps the project by demonstrating the distribution of outcomes specific to the family menu. It is useful to know how the breakdown of outcomes may differ between different menus because then LegalAid can assess where it may be under/over resourced, which menus are helping people at the highest rates, etc.

Issues faced (if any)

I did not work on this task with Vivienne and Sabrina prior to this round of presentations, so I am still becoming acquainted with the methods they are using to identify outcomes and the structure of the CAR dataset in general. I also had a hard time deciphering how in-depth to go with the outcomes specific to the family menu seeing as it does not have very many terminal nodes, so the breadth of outcomes quickly becomes overwhelming. I am afraid that I may have overgeneralized the outcomes too much for this presentation in an attempt to get succinct, easily understandable results, and we will probably need to delve deeper into outcomes for this menu for the future. I was also having a hard time catching all of the different fields in the dataset where an outcome could be indicated while still only counting distinct contact sessions for the percentages. Not all of the outcomes are spelled out in the terminal leg of a single contact session, so that made it difficult to always use grouping logic in my code to represent all of the potential outcomes.

Attempts to resolve issues (if any)

I tried multiple different ways of using case when logic for the outcomes, and I also attempted to create a variable that maps the entire journey of the call into one row (i.e., it would take each menu an ID visited and concatenate them into something like this: Family Menu -> Divorce Menu -> Simple Divorce Menu), but I was having trouble getting that to work in a way that would be meaningful for this objective.

Issues resolved (if any)

My approach eventually minimized NAs and showed a reasonable distribution of outcomes, but I still think that it is an overgeneralization that might not reflect the true spread of outcomes for the family menu.

Next steps

I am hoping that my teammates could help me create a more robust approach for the family menu due to all of the redirects to other menus and queues that make it more difficult to determine the outcomes.

References (Mention if you built up on someone else's work)

I used Vivienne and Sabrina's list of outcomes tied to termination reasons that they presented last round (on slide 6):

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/CallCategorization_Oct23_Group2.pdf

Presentation 2

Links to presentation(s) and code(s) on GitHub

Presentation:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/CallCategorization_Oct23_Group2.pdf

Code:

Data imported with these scripts:

CAR:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/CAR_data_import_R_CarolineCorr.R

All Calls:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/allcallsimport_Oct7_LoganRoever.R

Scripts that I worked on:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20integrity%20checks/AllInboundCallsNewStrategy_21Oct_CarolineCorr.R

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20integrity%20checks/AllInboundCalls_Upd_12Oct_CarolineCorr.R

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20integrity%20checks/CallDurationComparison_22Oct_CarolineCorr.R

What did you do?

I wrote the code to do the initial duration comparison between CAR and All Calls. I produced an initial plot that showed a huge difference between the two datasets, and I used these findings to guide our mentor meeting with Krish on Friday 10/17 so that we could better refine our approach, and Hailee implemented Krish's recommendations from that point. I also tweaked the code from last presentation's objective of comparing the number of inbound calls from both datasets. In doing so, I incorporated the 6 phone lines in All Calls and also attempted to filter by Edwin Rong's findings from presentation 1, but this strategy did not improve the disparity seen in the graphic.

How does it help the project?

My contributions help the project because in the case of the duration comparison, my findings provided a baseline for my teammates to improve upon after receiving Krish's feedback. I believe that my findings from my improved inbound call comparison graphic are helpful because they show us that the overall trends between both datasets are the same no matter which strategy we use to filter the All Calls data. This narrows the issue down to something within the CAR data that is causing the numbers to not line up.

Issues faced (if any)

I did not experience any technical issues for this presentation, but I am still struggling to identify what is causing the disparity between the number of inbound calls in CAR/All Calls.

Attempts to resolve issues (if any)

n/a

Issues resolved (if any)

I improved my inbound calls comparison by incorporating the six phone lines in All Calls that I did not consider before.

Next steps

We still need to identify why CAR seemingly has more inbound calls than All Calls. I think a thorough EDA of each field in CAR and its relationship with the overall number of calls could be revealing in this regard. However, if we can't identify a root cause, that would suggest that there is a deeper data quality issue at play here.

References (Mention if you built up on someone else's work)

I used Edwin Rong's strategy for filtering inbound calls from the All Calls dataset in my comparison. I also used the data import scripts that were linked above to synthesize the raw data into complete datasets.

Link to Edwin's documentation:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Documentation/Edwin_Rong_ProjectDocumentation.pdf

Presentation 1

Links to presentation(s) and code(s) on GitHub

Presentation:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Data%20integrity%20checks/CARAllCallsIntegrity_7Oct_EDATeam2.pdf

Code:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20integrity%20checks/AllInboundCalls_Upd_12Oct_CarolineCorr.R

What did you do?

Some of my teammates were facing difficulties getting the CAR data to read into their machine in our R project, so we decided to develop the code on my computer in the interest of time. I tweaked the CAR data import code from spring quarter of last year to include the CSV files, and then I created the graphic showing the monthly comparison for inbound calls between the CAR and All Calls dataset.

How does it help the project?

This helped the project by providing the basis for our analysis and visualization of data integrity. Using this comparison between both datasets as an indicator, we will be able to gauge how well the datasets align with one another and identify potential data quality issues.

Issues faced (if any)

We faced a lot of issues trying to read the data into R. I am not sure why, but I was the only one able to get the CAR data import code to run on my machine. I think the root cause may be our different operating systems (I am running Windows while they have Macs), but I'm not entirely sure. I was able to tweak the import code to include the more recent CSV files as well as the XLSX files with the help of ChatGPT. After solving this problem by deciding to just write all of the code on my computer, we thought it would be easier for Vivienne to upload her version of the All Calls dataset that she imported to her machine using the provided ipynb file to a shared Google drive for us to download locally, so that is why I do not have any import code for the All Calls dataset pushed to the GitHub repository.

After we had the data read in, we found that through our filtering (terminating calls, "Call Tower" PSTN, >0 seconds, 6 main lines, and unique Contact Session/Correlation IDs), the number of inbound calls between both datasets still does not match up, but the inclusion of the 6 lines definitely decreased the gap between the two. I believe once we gain a better understanding of the data we will be able to identify where this issue is coming from, especially since the gaps from month to month appear to have a similar magnitude, which doesn't seem like a coincidence.

Attempts to resolve issues (if any)

We luckily found some code from spring quarter of how to import the datasets in R, and we also consulted ChatGPT to help edit the code to include CSVs. We also decided to download the All Calls data locally from Vivienne's version in a shared Google drive instead of starting that R import code from scratch.

Issues resolved (if any)

We have now successfully loaded in both datasets, but we still have a long way to go with cleaning and understanding the data. We would like to gain a better understanding of columns such as "User Type" in all calls and what the number of inbound calls by calling hour (the objective initially set) is supposed to indicate for data quality. I understand that ideally, they should be the same by each hour, but what would this comparison tell us instead of using a different comparison category such as month or day of the week?

Next steps

It was brought to my attention that I did not take into account the 6 phone lines when filtering the all calls data, so moving forward I will be able to apply that and get a better idea of how inbound calls actually match up between both datasets.

Edit 10/12: I have now incorporated these 6 lines into my filtering, and the inbound calls are closer, but they still do not match up perfectly. Since they all seem to be off by a similar magnitude each month, I believe it is still a filtration issue that we will need to investigate further for our next presentation.

We also realized that it would have been more prudent to begin working on our goals before our meeting with Krish so that we could guide our questions better, so we will be sure to get started on our goals earlier for next presentation so we can ask better questions and get the help we need earlier.

References (Mention if you built up on someone else's work)

I used the R data import code from spring quarter to get started loading in the CAR data.

Link:
https://github.com/arvindkrishna87/STAT390_SP25_LegalAid/blob/main/Presentations/explore.R